Entity-Token World Models and Differentiable Predictive Control for Building-Agnostic HVAC Control

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Abstract

A single controller for the Heating, Ventilation and Air Conditioning (HVAC) systems of unseen buildings has to cope with a different number of zones, air handlers and thermal properties in every building and with climates ranging from subarctic winters to tropical summers. This work addresses the Smart Buildings Challenge with a policy that learns its own control logic, without deep reinforcement learning and without imitating an existing controller. Every zone, air handler and plant is represented as a token, and a transformer whose attention is biased only by the structural relation between tokens is used both as a world model of the building and as the policy, thereby making both independent of the building size. The policy is trained by Differentiable Predictive Control (DPC), backpropagating the competition reward through closed-loop rollouts of up to 24 hours inside the world model, which is refined in successive on-policy rounds. On the 30 official scenarios evaluated over 14 days, the final policy reduces the mean absolute step reward from 0.0591 for the shipped PID baseline to 0.0106, and it reaches 0.013 on the Kaggle public leaderboard. The experiments further indicate that the calibration of the world model on the reward terms matters more than its average temperature error.

1 Introduction

HVAC systems are still mostly operated by rule-based logic tuned by hand for each building [4, 13], since model predictive control and reinforcement learning, despite large savings on individual buildings, require a model or a training process specific to every building [11]. The Smart Buildings Challenge [7], built on the Smart Buildings Control Suite simulator [6], makes this obstacle explicit by asking for one policy that controls buildings never seen during development, with different floor plans, numbers of zones and air handlers, envelope materials and weather, and that is scored on a reward combining occupant comfort, energy cost and carbon emissions.

The approach presented here rests on two choices. First, the building is described as a set of entities (zones, air handlers, the hot-water system and a global token for weather and time), and the same transformer architecture, with attention biased by the structural relation between entities, is used for a world model of the building and for the policy, so that both apply to any building without modification. Second, the policy is not trained by trial and error on the simulator, nor by imitation of a baseline controller, but by DPC [5]: the competition reward is rebuilt from the world-model predictions and maximized by backpropagation through closed-loop rollouts, thereby letting the policy derive its own control logic from the physics captured by the world model, as outlined in Figure 1. The main contributions of this work are the following:

Smart Buildings Competition Track at 40th Conference on Neural Information Processing Systems (NeurIPS 2026).

- an entity-token world model with long thermal memory, refined in on-policy rounds, with a closed-loop temperature error of 0.82 K after 24 hours on unseen buildings;
- a DPC scheme with a 24-hour horizon and a command filter inside the rollouts, which reaches 0.0106 locally and 0.013 on Kaggle, against 0.0591 for the shipped baseline;
- an analysis showing that the calibration of the world model on the reward terms, rather than its average error, determines the quality of the policy.

2 Related Works

The Smart Buildings Challenge [7] is the first benchmark in which a single HVAC policy is evaluated on unseen buildings generated at scale with the simulator of the Smart Buildings Control Suite [6], whereas BOPTEST [2] provides detailed individual test cases. Model Predictive Control (MPC) is the reference for advanced building control [4], although its dependence on a model identified for every building limits its scalability. Deep reinforcement learning has been applied to HVAC control since [14], with sample efficiency, safety and transfer to new buildings as its main obstacles [13, 11].

Differentiable MPC [1] embeds an optimization problem in a policy and has been applied to HVAC control with a linear model [3], whereas DPC [5], adopted in this work, trains an explicit neural policy offline by differentiating a control objective through a model of the system. Learned world models [8, 9] allow policies to be trained in imagination, at the risk of exploiting model errors [10]; short-horizon actor-critic methods such as SHAC [15] address long horizons with a learned value function, whereas the present work relies on an accurate world model and on gradient damping to differentiate directly through horizons of 24 hours. The entity-token architecture follows the transformer [12], and the attention bias depending on the relation between two nodes is inspired by Graphormer [16].

3 Methods

3.1 Problem Setting

At every 5-minute step t, the policy receives the observation o_t of a building with V zones and K air handlers, which contains the measurements of every zone, air handler and hot-water system, the comfort-mode flags, the occupancy and the time, and it returns the action a_t, made of the damper and reheat commands of every zone, the run state, supply-air temperature, pressure and recirculation of every air handler and the run state, water temperature and pressure of the hot-water system. The step reward combines occupant comfort, energy cost and carbon emissions as follows:

$$r_t = w_p\, R_t - w_e\, \tilde{E}_t - w_c\, \tilde{C}_t, \tag{1}$$

where w_p, w_e and w_c are the weights fixed by the competition, $\tilde{E}_t$ and $\tilde{C}_t$ are the energy cost and the carbon emissions of electricity and natural gas, each normalized by its value at the maximum consumption rates, and $R_t \le 0$ is the productivity regret of the occupants, defined as:

$$R_t = -\frac{p_{\max}}{p_{\max} - p_{\min}} \frac{\sum_v n_{v,t}\, \varphi(\delta_{v,t})}{\sum_v n_{v,t}}, \qquad \delta_{v,t} = \max\left(T_{v,t} - T^{\mathrm{c}}_{v,t},\; T^{\mathrm{h}}_{v,t} - T_{v,t},\; 0\right), \tag{2}$$

where $n_{v,t}$ is the number of occupants of zone v, $T_{v,t}$, $T^{\mathrm{h}}_{v,t}$ and $T^{\mathrm{c}}_{v,t}$ are its air temperature and its heating and cooling setpoints, $p_{\max}$ and $p_{\min}$ are the maximum and minimum productivity per person-hour, and the shortfall $\varphi(\delta) = \delta/\delta_0$ for $\delta \le \delta_0$ and $\varphi(\delta) = 1 + \left((\delta - \delta_0)/\delta_0\right)^2$ beyond the reference deviation δ_0 grows linearly and then quadratically with the distance from the comfort band. The score of a policy is the mean absolute step reward $S = \frac{1}{N}\sum_t |r_t|$ over the N steps of all scenarios, so that lower is better.

3.2 Entity-Token Architecture

As can be observed in Figure 1(b), a building is represented by $N_b = V + K + 2$ tokens, one per zone, one per air handler, one for the hot-water system and one global token. The input vector in_i of a zone concatenates its normalized observations, its last three temperature changes, its previous commands and seven static descriptors of the floor plan (floor area, exterior-wall exposure, floor

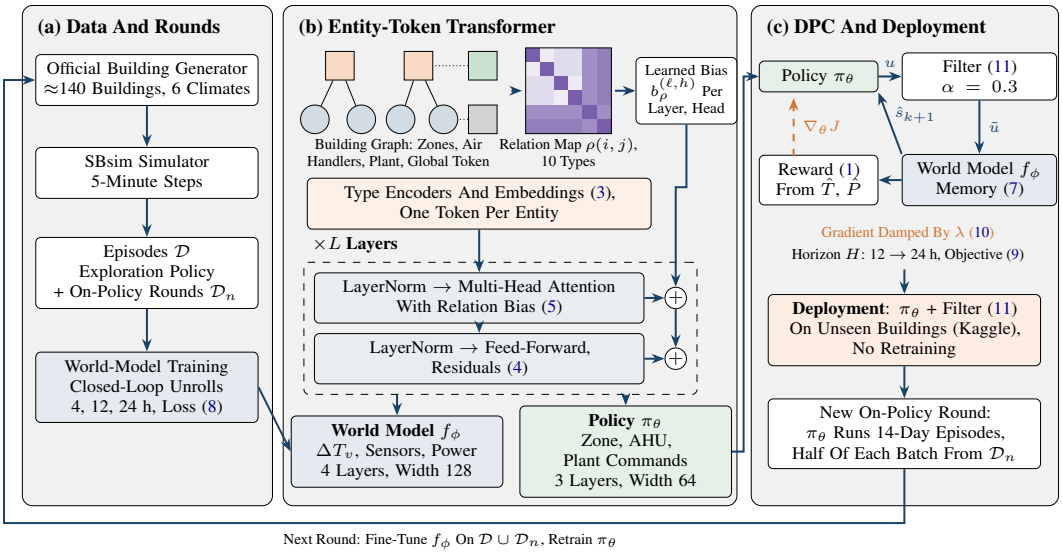


Figure 1: Overview of the method: (a) data collection on generated buildings and on-policy rounds; (b) entity-token transformer shared by the world model and the policy (same structure, separate weights), with Graphormer-style attention biases given by the relation between two entities; (c) DPC training of the policy through the world model and deployment on unseen buildings.

position and number of zones on the same air handler); air-handler and hot-water tokens concatenate their sensors, including the outdoor air temperature, with their previous commands, while the global token carries the comfort-mode flags, the occupancy and the time, together with the hours to the next scheduled comfort period for the policy. Every input vector is first embedded as follows:

$$x_i^{(0)} = \mathrm{MLP}_{c(i)}(\mathrm{in}_i) + e_{c(i)}, \tag{3}$$

where $c(i) \in \{\mathrm{zone}, \mathrm{ahu}, \mathrm{hws}, \mathrm{glob}\}$ is the type of token i, MLP_c is a two-layer encoder specific to that type and $e_c \in \mathbb{R}^d$ is a learned type embedding of the model width d. The tokens are then processed jointly by L pre-norm transformer layers [12], each made of an attention block and a feed-forward block with residual connections:

$$\bar{x}_i^{(\ell)} = x_i^{(\ell-1)} + \sum_{h=1}^{n_h} W_O^{(\ell,h)} \sum_{j=1}^{N_b} A_{ij}^{(\ell,h)}\, v_j^{(\ell,h)}, \qquad x_i^{(\ell)} = \bar{x}_i^{(\ell)} + \mathrm{FFN}^{(\ell)}\big(\mathrm{LN}(\bar{x}_i^{(\ell)})\big), \tag{4}$$

where LN is layer normalization, $\mathrm{FFN}^{(\ell)}$ a two-layer perceptron with GELU activation, n_h the number of heads, $v_j^{(\ell,h)}$ the value vector of token j and $W_O^{(\ell,h)}$ the output projection of head h. The only structural information about the building enters through the attention weights $A_{ij}^{(\ell,h)}$, in the spirit of Graphormer [16]:

$$A_{ij}^{(\ell,h)} = \mathrm{softmax}_j\left(\frac{q_i^{(\ell,h)} \cdot k_j^{(\ell,h)}}{\sqrt{d/n_h}} + b_{\rho(i,j)}^{(\ell,h)}\right), \tag{5}$$

where $q_i^{(\ell,h)}$ and $k_j^{(\ell,h)}$ are the query and key vectors computed from $\mathrm{LN}(x^{(\ell-1)})$, $\rho(i,j)$ is one of ten relations between two tokens, determined only by the map of zones to air handlers (for instance, two zones on the same air handler, or a zone and its own air handler), and $b_\rho^{(\ell,h)}$ is a learned scalar per relation, layer and head, initialized to zero. Since no parameter depends on V or K, the same weights apply to every building. Separate linear heads read the output tokens $x_i^{(L)}$: the continuous commands of the policy pass through a hyperbolic tangent, while its on/off commands use a hard threshold in the forward pass and the gradient of the underlying sigmoid in the backward pass (straight-through estimator).

3.3 World Model

Given the state s_t, that is, the token inputs above, the action a_t and the exogenous inputs x_t read from the episode (weather, setpoints, occupancy, tariffs and carbon intensities), the world model f_ϕ predicts the next actuator readbacks of every zone, the next sensor values of every air handler and of the hot-water system, the zone temperatures and the power of the building, the last two as follows:

$$\begin{aligned} \hat{T}_{v,t+1} &= \hat{T}_{v,t} + \sigma_{\Delta T}\, y^{T}_{v,t}, \\ \hat{P}^{\mathrm{el}}_{t} &= P^{\mathrm{el}}_{0}\Big(\textstyle\sum_k \mathrm{softplus}(y^{\mathrm{el}}_{k,t}) + \mathrm{softplus}(y^{\mathrm{el}}_{\mathrm{hws},t})\Big), \\ \hat{P}^{\mathrm{gas}}_{t} &= P^{\mathrm{gas}}_{0}\Big(\textstyle\sum_v \mathrm{softplus}(y^{\mathrm{gas}}_{v,t}) + \mathrm{softplus}(y^{\mathrm{gas}}_{\mathrm{hws},t})\Big), \end{aligned} \tag{6}$$

where $y^{T}_{v,t}$ is the output of the zone head for the normalized temperature change, $\sigma_{\Delta T}$ the standard deviation of the 5-minute temperature changes in the training data, y^{el} and y^{gas} the outputs of the air-handler, zone and hot-water heads for the electric power of fans and pumps and for the gas power of reheat and boiler, and $P^{\mathrm{el}}_0 = 80$ kW and $P^{\mathrm{gas}}_0 = 205$ kW scale constants; the softplus keeps every contribution non-negative, and the reward (1) is rebuilt exactly from $\hat{T}$ and $\hat{P}$. Since walls and interior mass store heat that is not observed, each zone token is augmented by a long memory of exponential averages:

$$m^{(z,\tau)}_{v,t+1} = \beta_\tau\, m^{(z,\tau)}_{v,t} + (1-\beta_\tau)\, z_{v,t+1}, \qquad \beta_\tau = e^{-1/\tau}, \tag{7}$$

where the signal z is the zone temperature, with time constants τ of 12, 72 and 288 steps (1, 6 and 24 hours), the reheat and damper commands, with 6 and 24 hours, and the outdoor air temperature, with 6 and 24 hours; during a rollout the averages are updated from the predicted temperatures $\hat{T}$.

The model is trained on closed-loop unrolls of $U \in \{48, 144, 288\}$ steps (4, 12 and 24 hours), in which its own predictions are fed back while the logged actions are replayed, by minimizing the following loss:

$$\mathcal{L}(\phi) = \frac{1}{U}\sum_{k=1}^{U}\Big(\frac{\sum_v \omega_{v,k}\,(\hat{T}_{v,k} - T_{v,k})^2}{\sigma^2_{\Delta T}\sum_v \omega_{v,k}} + w_R\Big(\frac{\hat{R}_k - R_k}{\sigma_R}\Big)^2 + w_s\,\mathcal{L}^{\mathrm{sens}}_k + w_P\,\big\|\hat{P}_k - P_k\big\|^2\Big), \tag{8}$$

where $\omega_{v,k} = 1 + 3\cdot\mathbb{I}[\,n_{v,k} > 0$ and $T_{v,k}$ within 1 K of or outside the comfort band$\,]$ concentrates the accuracy where the reward is sensitive to temperature errors, the second term matches the regret (2) directly, with $\sigma_R = 0.1$, so that the error is measured in the units of the DPC objective, $\mathcal{L}^{\mathrm{sens}}_k$ is the mean squared error on the normalized readbacks and sensors, and w_R, w_s and w_P are the weights in Table 1; unrolls longer than 4 hours are recomputed in the backward pass (gradient checkpointing), so that memory does not grow with U.

The training data come from about 140 buildings produced by the official generator, with seeds disjoint from all evaluation sets, under six weather stations (Atlanta, Denver, Fairbanks, Minneapolis, Phoenix and Singapore). The first episodes come from an exploration policy switching between the shipped PID controller, its noisy version, randomly held setpoints and extreme plant states; afterwards, policy and world model alternate in on-policy rounds (Algorithm 1), in which the current policy runs one long episode per building, partly with action noise, and the world model is fine-tuned with half of each batch from the new episodes, thereby correcting it in the states that the policy visits. Five successive world models are compared in Section 4: WM-1 and WM-2 come from the exploration data and the first rounds, WM-3 introduces the memory (7) and the 24-hour unrolls, WM-4 adds a round of the first filtered policy, and WM-5 a round in which a quarter of the segments hold the plant setpoints at random levels.

3.4 Differentiable Predictive Control

The policy is trained by DPC [5], as detailed in the inner loop of Algorithm 1 and in Figure 1(c). From a batch of true start states s_{t_0} drawn from the collected episodes, the policy acts on the predicted states, the world model predicts the next state with the exogenous inputs of the episode and the reward is rebuilt from the prediction, so that the following objective is maximized by backpropagation through the whole closed loop:

$$J(\theta) = \mathbb{E}_{s_{t_0}}\Big[\frac{1}{H}\sum_{k=0}^{H-1}\gamma^k\,\hat{r}_{t_0+k}\Big], \qquad \hat{r}_t = \kappa\, w_p \hat{R}_t - w_e\hat{\tilde{E}}_t - w_c\hat{\tilde{C}}_t, \tag{9}$$

Algorithm 1 On-policy rounds of world-model training and DPC training of the policy.

$\mathcal{D} \leftarrow$ exploration episodes on the generated buildings; train f_ϕ on $\mathcal{D}$ with (8)
for round $n = 1, 2, \ldots$ **do**
  for iteration $i = 1, \ldots, I$ **do** ▷ DPC, warm-started from the previous policy
    $H \leftarrow$ curriculum horizon; sample an episode and B start indices t_0; $\hat{s}_{t_0} \leftarrow s_{t_0}$
    for $k = 0, \ldots, H-1$ **do**
      $\tilde{u} \leftarrow \alpha\, \pi_\theta(\hat{s}_{t_0+k}) + (1-\alpha)\, \tilde{u}$ (11); $\hat{s}_{t_0+k+1} \leftarrow f_\phi(\hat{s}_{t_0+k}, \tilde{u}, x_{t_0+k})$; memory (7)
      $\hat{r}_{t_0+k} \leftarrow$ reward rebuilt from $\hat{T}$ and $\hat{P}$ (9); damp the state gradient (10)
    end for
    $\theta \leftarrow \mathrm{AdamW}(\theta, -\nabla_\theta J)$ with clipped gradient norm, unless the loss is not finite
  end for
  $\mathcal{D}_n \leftarrow$ one 14-day episode per generated building with π_θ, partly with action noise
  fine-tune f_ϕ on $\mathcal{D} \cup \mathcal{D}_n$, half of every batch from $\mathcal{D}_n$; $\mathcal{D} \leftarrow \mathcal{D} \cup \mathcal{D}_n$
end for

where H is the horizon, γ the discount factor, $\hat{r}_t$ the reward (1) computed from the predicted temperatures and power of (6), and $\kappa > 1$ a factor on the comfort term that counteracts the tendency of the policy to trade comfort for energy inside the world model. Three mechanisms make the optimization over horizons of hundreds of steps stable and transferable to the simulator. First, the gradient carried by the predicted state from one step to the previous one is damped without changing its value:

$$\hat{s}_{k+1} \leftarrow \mathrm{sg}(\hat{s}_{k+1}) + \lambda\big(\hat{s}_{k+1} - \mathrm{sg}(\hat{s}_{k+1})\big), \tag{10}$$

where $\mathrm{sg}(\cdot)$ is the stop-gradient operator and $\lambda < 1$ the damping factor, so that the credit assigned to an action by a reward h steps later is weighted by λ^h and the backpropagation does not explode. Second, since the policy otherwise reacts to world-model errors with fast oscillating commands, which the real plant does not reward, every continuous command is filtered both inside the rollouts and at deployment:

$$\tilde{u}_t = \alpha\, u_t + (1-\alpha)\, \tilde{u}_{t-1}, \tag{11}$$

where u_t is the command produced by the policy, $\tilde{u}_t$ the command applied to the building and $\alpha \in (0, 1]$ the filter coefficient; on/off commands are not filtered, and the policy receives $\tilde{u}_{t-1}$ as its previous action, exactly as in the building. Third, the horizon follows a curriculum from 12 to 24 hours, which is long enough for the policy to experience the effect of preheating before occupancy and of letting the building drift at night and over the weekend. The start states exclude the generated buildings larger than the official size range, and every policy is warm-started from the policy of the previous round. The architecture and training settings of both networks are listed in Table 1.

Table 1: Architecture and training settings of the world model and of the final policy.

World Model	
Layers, width, heads, feed-forward	4, 128, 4, 256
Parameters	0.61 M
Unroll lengths U [steps]	48, 144, 288
Loss weights w_R, w_s, w_P	5, 0.3, 10
Memory time constants τ [steps]	12, 72, 288
Batch, steps per round	16 unrolls, 1000
Optimizer, learning rate	AdamW, $5 \cdot 10^{-5}$

Policy And DPC	
Layers, width, heads, feed-forward	3, 64, 4, 128
Parameters	0.12 M
Horizon H (curriculum) [steps]	144, 288
Discount γ, damping λ	0.995, 0.99
Comfort factor κ, filter α	1.8, 0.3
Batch B, iterations I	24 starts, 400
Optimizer, learning rate	AdamW, $3 \cdot 10^{-4}$

4 Results

4.1 Evaluation Protocol

Policies are evaluated in the simulator on the 30 official scenarios (six buildings under five weather stations), none of which is used for training, over 14 days, so that every episode includes a weekend and a Monday start-up. The local 14-day score proved to be a reliable predictor of the Kaggle evaluation, with a ratio between the two of 1.23, 1.29 and 1.23 for the three policies submitted, against 1.39 and 1.48 for the 7-day protocol.

4.2 Overall Performance

The progression of the method is summarized in Figure 2 and its effect on the individual scenarios in Figure 3.

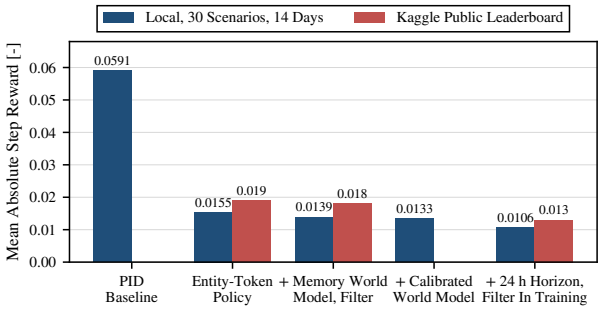


Figure 2: Mean absolute step reward of the shipped PID baseline and of the successive stages of the method, on the 30 official scenarios over 14 days and on the Kaggle public leaderboard.

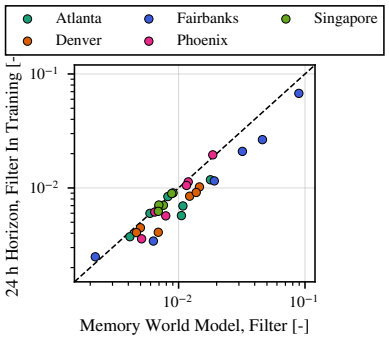


Figure 3: Score of each official scenario (14 days): filtered policy on WM-3 against the final policy; points below the diagonal are improvements.

As can be observed in Figure 2, the first entity-token policy trained by DPC already reduces the score of the PID baseline by 74% (from 0.0591 to 0.0155), and every following stage brings a further reduction, down to 0.0106 for the final policy, that is, 0.013 on the Kaggle public leaderboard against 0.019 for the first submission.

As can be observed in Figure 3, the final policy is better in 23 of the 30 scenarios, with the largest reductions in the cold climates (Fairbanks 1 from 0.089 to 0.068, Fairbanks 8 from 0.046 to 0.027). Figure 4 shows two heating scenarios in very different climates, Fairbanks 1 in January (outdoor air below $-35\,^{\circ}\mathrm{C}$) and Denver 8 in November.

As can be observed in Figure 4, both policies switch the plant off at night and over the weekend, and the difference lies in the heat delivered during occupancy: in Fairbanks 1 the final policy heats more in the second half of each working day, thereby keeping the coldest zones closer to the setpoint with 11% more gas (weekly mean step reward from -0.0650 to -0.0548), and in Denver 8 its coldest zones reach the setpoint sooner after the morning start (from -0.0174 to -0.0121). Over the 30 scenarios the comfort term is more than halved (from 0.0065 to 0.0026), while energy and carbon grow slightly (from 0.0074 to 0.0080).

4.3 World-Model Quality and Policy Quality

Table 2 relates the diagnostics of the five world models, computed from 1080 start states of 45 clean episodes, to the policies trained on each model.

Table 2: World models (Section 3.3) after 24 hours: temperature error and bias, ratio of predicted to true comfort and energy terms, and 14-day score of the policies trained on them.

World Model	Replay MAE [K]	Closed-Loop MAE [K]	Closed-Loop Bias [K]	Comfort Pred./True	Energy Pred./True	Policy (Horizon) 14-Day Score [-]
WM-1	1.38	1.13	+0.43	0.96	0.95	Entity-Token (6–12 h): 0.0155
WM-2	1.50	1.35	+1.07	0.92	1.09	–
WM-3	0.87	0.87	−0.18	1.22	1.22	Filter (6–12 h): 0.0139
WM-4	0.89	0.82	+0.06	1.00	1.10	Filter (6–12 h): 0.0133 Filter In Training (24 h): **0.0106**
WM-5	0.78	0.79	−0.01	1.07	1.20	Filter In Training (24 h): 0.0121

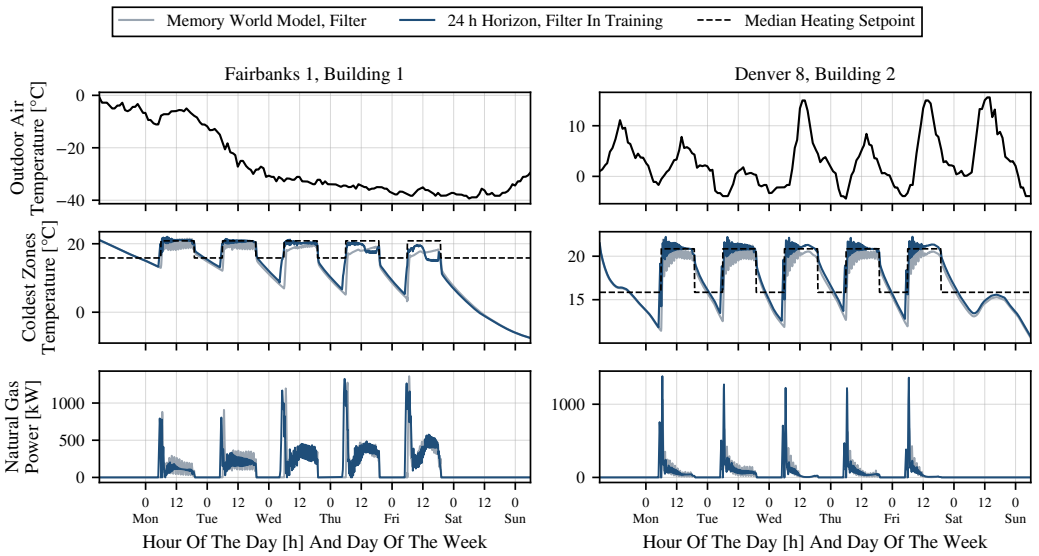


Figure 4: First week of Fairbanks 1 (official building 1, from 1 January) and of Denver 8 (official building 2, from 5 November): outdoor air temperature, 10th percentile of the zone temperatures and natural gas power of the filtered policy trained on WM-3 and of the final policy.

As reported in Table 2, the memory of WM-3 reduced the 24-hour error by about 35% with respect to WM-2, and the round of WM-4 removed the temperature bias and the overestimation of discomfort (from 1.22 to 1.00). WM-5 has the lowest error of all, yet the same recipe yields a worse policy on it than on WM-4 (0.0121 against 0.0106): WM-5 overestimates the comfort term by 7% and the energy term by 20%, and its policy spends more energy than needed, especially in warm climates. This can primarily be attributed to the DPC gradient, which follows the predicted reward rather than the temperature, so that a systematic bias on the reward terms steers the policy, whereas random temperature errors largely average out.

4.4 Long Horizon And Command Filter

The largest gain in Figure 2 (from 0.0133 to 0.0106) combines the 24-hour horizon with the filter inside the training rollouts, two elements effective only together and on an accurate world model: on WM-4, the filter with a 6 to 12-hour horizon yields 0.0182, since short rollouts do not reveal the benefit of acting earlier with slower commands. On WM-2, whose 24-hour error is 1.35 K (Table 2), the long horizon instead degraded the 7-day score from 0.0195 to 0.0328, since the policy exploited the accumulated errors.

5 Conclusion

This work has shown that a single policy for unseen buildings can be obtained without deep reinforcement learning and without imitation, by differentiating the competition reward through closed-loop rollouts of an entity-token world model, thereby reducing the score of the shipped PID baseline by 82% and reaching 0.013 on the Kaggle public leaderboard. The main driver of performance is the world model, in particular its calibration on the reward terms and its accuracy over the horizon. The next steps are the variance between training seeds, ensembles of world models and planning at deployment.

Acknowledgments. This work was carried out within the CREALab project, funded by the Italian Ministry of the Environment and Energy Security (MASE) under the Mission Innovation 2.0 program (project code MI_FAE_00326).

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